No:7 Examine Network Address Translation using NAT in CISCO Packet Tracer

Aim:
To configure and examine Network Address Translation (NAT) in Cisco Packet Tracer and verify communication b/w a private network and an external network.

Requirements:
Cisco Packet Tracer
One Router (Cisco 2911)
Two PCs
One server
copper straight-through cables

Procedure:
1. Place one router, two PCs, and one server in the workspace.
2. Connect the PCs to the router's inside network and the server to the outside network.
3. Configure the IP addresses of the PC's router interfaces, and server.
4. Configure NAT on the router to translate private IP addresses into a public IP address
5. mark the inside and outside interfaces.
6. Test connectivity using the ping command.
7. Verify NAT translations using the show ip nat translations command.

IP Address configuration:

Device	Interface	IP Address
PC-0	NIC	192.168.1.2/24
PC-1	NIC	192.168.1.3/24
Router (Inside)	G0/0	192.168.1.1/24
Router (outside)	G0/1	203.0.113.1/24
Server	NIC	203.0.113.2/24

Router configuration:

Router> enable

Router# configure terminal

Router(config)# access-list 1 permit 192.168.1.0

Router(config)# interface gigabit Ethernet 0/0

Router(config-if)# ip nat inside

Router(config-if)# no shut down

Router(config-if)# exit

Router(config)# interface gigabit Ethernet 0/1

Router(config-if)# ip nat outside

Router(config-if)# exit

Router(config)# ip nat inside list 1

Router(config)# end

Router# write memory

Verification:

From PC-0, open Desktop - command Prompt and enter

ping 203.0.113.2

on the router. execute

show ip nat translations

show ip nat statistics

Result:

NAT was configured and examined in CiscoPacket Tracer successfully.